
☒ FINAL REVISION SHEET – (UNIT-1 + UNIT-2 + UNIT-3)

Cloud Computing • Virtualization • SOA • OpenStack • Docker • CI Principles

☒ CLOUD COMPUTING – CORE POINTS

Cloud Characteristics

- On-demand
- Elasticity (auto-scale up/down)
- Resource pooling
- Measured service (pay-per-use)
- Broad network access

Cloud Models

- SaaS: Ready apps (Gmail, Salesforce)

- **PaaS:** Developer platform (Heroku, App Engine)
- **IaaS:** Virtual machines, networks (AWS EC2, GCP Compute Engine)

Deployment Models

- **Public:** Shared infrastructure (AWS, Azure)
 - **Private:** Single organization
 - **Hybrid:** Mix of public + private
 - **Community:** Shared by similar orgs (banks, gov)
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☒ VIRTUALIZATION – ESSENTIAL POINTS

Hypervisors

- **Type-1 (Bare Metal):** Runs directly on hardware
Examples: VMware ESXi, Hyper-V
- **Type-2 (Hosted):** Runs on OS
Examples: VirtualBox, VMware Workstation

Virtualization Techniques

- **Full Virtualization:**
 - Guest OS unmodified
 - Binary translation
- **Para-Virtualization:**
 - Requires modified guest OS
 - Hypercalls
- **Hardware-Assisted Virtualization:**
 - Intel VT-x / AMD-V support

Virtualization Benefits

- Isolation
- Portability
- Easy migration
- Efficient resource usage

☒ SOA (Service-Oriented Architecture)

Key Concepts

- Services are reusable, loosely coupled, independent
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Communication via **standard protocols** (HTTP/SOAP/REST)

- Platform independent
- Easy integration between systems

Advantages

- Reusability
- Maintainability
- Scalability
- Loose coupling

Disadvantages

- High overhead
- Security challenges
- Complex service management

- **Domain:** Business problem
- **Ubiquitous Language:** Same terms for devs + business
- **Bounded Context:** Separate business areas
- **Entity:** Identified by ID
- **Value Object:** Immutable, no ID
- **Aggregate:** Group of entities/values
- **Repository:** Saves aggregates
- **Domain Event:** Something important happened
- **Anti-Corruption Layer:** Protects from external models
- **Application Layer:** Orchestration
- **Infrastructure Layer:** DB, APIs, I/O

☒ OPENSTACK — ALL COMPONENTS IN 1 PLACE

Component	Purpose
Nova	Compute (create/manage VMs)
Neutron	Networking (IP, subnets, routers)
Swift	Object storage
Cinder	Block storage
Keystone	Identity + token authentication
Glance	VM images (store/retrieve)
Horizon	Dashboard UI
Ceilometer	Metering/billing
Heat	Orchestration & auto-scaling

Most asked questions

- Swift = object storage
- Cinder = persistent block storage
- Glance = images
- Nova = VMs
- Keystone = auth
- Horizon = dashboard

☒ DOCKER — MUST-KNOW POINTS

What is Docker?

Containerization platform: run apps anywhere, same environment.

Docker Architecture

- Docker Client: You type commands
- Docker Daemon: Does actual work (build, run, stop containers)
- Docker Registry: Stores images (Docker Hub)

Container = running instance of an image

Docker Networks

- bridge (default): Local container-to-container comm
- host: Uses host network directly
- none: No network
- overlay: Multi-host network (Swarm)

Useful Commands

```
docker ps
docker images
docker network ls
docker run -itd nginx
docker exec -it <id> sh
docker build -t app .
```

☒ CONTAINER ORCHESTRATION (No Kubernetes Details)

Why orchestration?

- Auto-scaling
- Auto-restart
- Load balancing
- Deployment automation
- Health checks

Popular Tools:

- Docker Swarm (comes with Docker)
- Apache Mesos
- (Kubernetes removed as you requested)

☒ CI (Continuous Integration) – EXAM

HIGHLIGHTS

- All code in **one repository**
- **Automate builds**
- **Automate tests**
- Every commit → build
- Fix broken builds immediately
- Keep builds fast
- Test in production-like environment
- Automate deployment
- Use static analysis + code coverage

☒ 10 EXPRESS MCQs FOR FINAL REVISION

1. Swift provides:

☒ Object Storage

2. Docker's default network:

☒ bridge

3. Heat is used for:

- ☒ Orchestration

4. Para-virtualization requires:

- ☒ Modified guest OS

5. Cinder provides:

- ☒ Block storage

6. Keystone is responsible for:

- ☒ Authentication / tokens

7. Docker Client talks to Daemon using:

- ☒ REST API / Unix socket

8. Entity in DDD is identified by:

- ☒ ID

9. SaaS provides:

- ☒ Ready-to-use applications

10. SOA services are:

- ☒ Loosely coupled & reusable